

CRESTOMERE AGCM, AB, Canada

WMO: 713800

Lat: 52.7328N

Lon: 113.9028W

Elev: 855

StdP: 91.47

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.8	-27.3	-34.1	0.2	-30.8	-30.7	0.2	-27.2	13.4	-4.0	12.0	-5.2	1.7	150	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	27.5	16.9	26.0	16.0	24.4	15.5	18.8	24.9	17.7	23.7	16.7	22.6	3.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	13.1	21.8	15.3	12.1	20.6	14.3	11.2	19.4	57.0	25.0	53.1	23.9	50.0	22.6	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	9.5	8.3	-36.0	30.8	2.8	1.8	-38.0	32.1	-39.7	33.1	-41.2	34.1	-43.2	35.4		
			-36.1	21.3	2.8	1.8	-38.1	22.6	-39.7	23.6	-41.3	24.6	-43.3	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-8.3	-10.1	-4.0	3.2	10.0	13.8	16.2	15.3	10.7	4.0	-4.1	-9.9
	DBStd	11.35	8.26	8.87	8.02	5.15	4.32	2.95	2.63	3.01	4.46	4.81	7.10	7.82
	HDD10.0	3108	567	562	433	208	56	5	0	2	44	193	422	616
	HDD18.3	5559	826	795	692	453	260	138	74	99	231	445	672	874
	CDD10.0	611	0	0	0	5	54	119	194	168	64	7	0	0
	CDD18.3	21	0	0	0	0	1	2	10	7	1	0	0	0
	CDH23.3	571	0	0	0	2	54	61	206	164	82	3	0	0
	CDH26.7	91	0	0	0	0	5	6	36	28	16	0	0	0
Wind	WSAvg	3.4	3.6	3.5	3.4	4.1	4.1	3.6	3.0	2.7	3.0	3.5	3.2	3.4
Precipitation	PrecAvg	489	24	18	20	30	56	92	94	58	36	24	23	14
	PrecMax	607	76	36	45	61	116	146	172	107	80	71	59	30
	PrecMin	340	3	1	1	6	19	26	30	20	1	3	4	4
	PrecStd	75	17	10	12	15	24	34	38	20	21	18	15	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	8.6	13.4	22.5	27.4	27.3	29.9	29.6	28.7	21.7	15.4	7.7
		MCWB	3.3	3.3	6.4	10.5	13.2	16.3	17.6	17.6	15.4	10.9	7.9	2.5
	2%	DB	7.1	6.0	10.0	18.9	24.8	25.1	27.4	26.9	25.9	17.8	10.7	5.3
		MCWB	2.9	2.1	4.3	9.0	12.4	14.9	18.4	17.2	14.7	9.7	5.1	1.4
	5%	DB	5.3	3.8	7.7	14.7	22.3	23.3	25.7	25.1	22.9	14.9	7.1	3.1
		MCWB	1.8	0.6	3.2	6.3	11.4	14.3	17.3	17.0	13.8	8.1	3.0	-0.1
	10%	DB	3.1	2.0	5.4	12.0	19.7	21.4	23.7	23.2	20.0	12.2	5.0	0.9
		MCWB	0.2	-0.7	1.8	5.1	10.2	13.4	16.4	16.3	12.6	6.6	1.8	-1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.8	6.7	10.7	14.2	17.7	21.4	20.3	16.7	12.2	7.7	3.2
		MCDB	7.9	7.8	12.7	21.9	25.2	23.7	26.8	25.6	24.9	20.8	14.3	6.9
	2%	WB	3.0	2.1	4.8	8.7	12.9	16.3	19.2	18.7	15.5	10.2	5.2	1.7
		MCDB	6.8	5.5	9.4	18.4	21.8	22.9	25.0	25.0	23.7	17.1	9.5	5.1
	5%	WB	1.9	0.9	3.2	7.0	11.9	15.2	18.0	17.6	14.2	8.5	3.5	0.1
		MCDB	5.2	3.6	7.4	13.4	20.9	21.3	24.1	23.6	21.7	13.9	6.6	2.9
	10%	WB	0.4	-0.5	2.0	5.5	10.8	14.2	17.0	16.6	12.9	7.0	2.0	-1.7
		MCDB	2.9	1.9	5.0	11.2	18.8	20.0	22.8	22.5	19.8	11.6	4.6	0.6
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.9	10.8	11.6	13.7	12.2	12.8	13.2	14.0	12.0	10.1	10.1
		MCDBR	10.7	10.9	12.4	17.5	19.0	15.9	16.6	17.1	20.3	16.8	12.4	11.0
		MCWBR	7.6	7.8	7.7	9.2	8.5	7.4	8.6	9.0	10.1	9.5	8.0	8.1
	5% WB	MCDBR	10.6	10.3	12.1	16.9	17.4	13.7	15.3	15.8	18.3	16.1	11.8	11.2
		MCWBR	7.8	7.7	7.9	9.1	8.0	7.0	8.3	8.7	9.5	9.4	8.1	8.4
Clear-Sky Solar Irradiance	taub		0.259	0.262	0.299	0.324	0.338	0.345	0.348	0.358	0.309	0.282	0.252	0.242
	taud		2.312	2.390	2.390	2.383	2.395	2.398	2.422	2.360	2.483	2.541	2.470	2.317
	Ebn at Noon		747	871	898	912	912	906	896	865	878	841	769	702
	Edh at Noon		59	74	92	106	110	112	107	108	84	64	50	48
All-Sky Solar Radiation	RadAvg		1.03	2.04	3.47	4.66	5.62	5.95	6.22	5.13	3.56	2.12	1.08	0.77
	RadStd		0.09	0.16	0.24	0.42	0.38	0.40	0.34	0.35	0.38	0.24	0.10	0.07

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page